

Week 3 In-class Assignment

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```
library(tidyverse)
library(knitr)
library(kableExtra)
library(dplyr)
library(janitor)
```

Prepare World Bank data

1- Reading the data. First we read in maternal mortality data, then changing to long format.

```
# Read in maternal mortality data
matmor <- read.csv("../data/raw/maternal_mortality.csv", header = TRUE)

# Change wide to long format
matmor_long <- matmor |>
  pivot_longer(cols = starts_with("X"),
               names_to = "year",
               names_prefix = "X", # removes X from year column
               values_to = "maternal_mortality") |>
  mutate(year = as.numeric(year)) |> # change year to numeric
  select(iso, year, maternal_mortality)
```

2- Creating a function that can be used to convert all data sets we have into long formats.

```
mortality_fun <- function(data, varname) {

  data |>
    pivot_longer(
      cols = starts_with("X"),
      names_to = "year",
      names_prefix = "X",
      values_to = varname
    ) |>
```

```

    mutate(year = as.numeric(year)) |>
    select(iso, year, all_of(varname))
}

```

```

infmor <- read.csv("../data/raw/infant_mortality.csv", header = TRUE)

infmor_long <- mortality_fun(
  data = infmor,
  varname = "infant_mortality"
)

neomor <- read.csv("../data/raw/neonatal_mortality.csv"), header = TRUE)

neomor_long <- mortality_fun(
  data = neomor,
  varname = "neonatal_mortality"
)

un5mor <- read.csv("../data/raw/under5_mortality.csv"), header = TRUE)

un5mor_long <- mortality_fun(
  data = un5mor,
  varname = "under5_mortality"
)

```

Prepare disaster data

1- Reading the disaster data.

```

disaster <- read.csv("../data/raw/disaster.csv", header = TRUE)

```

2- Cleaning the variable names, then including only data for 2000-2019, and finally creating the dummy variable that counts droughts and earthquakes.

```

disaster_clean <- disaster |>
  clean_names() |>
  filter(
    year >= 2000,
    year <= 2019,
    disaster_type %in% c("Earthquake", "Drought")
  ) |>
  select(year, iso, disaster_type) |>
  mutate(
    earthquake = if_else(disaster_type == "Earthquake", 1, 0),

```

```

    drought = if_else(disaster_type == "Drought", 1, 0)
  ) |>
  group_by(year,iso) |>
  summarise(
    earthquake = max(earthquake),
    drought = max(drought),
    .groups = "drop"
  )

```

Prepare conflict data

1- Reading the conflict data set.

```

conflict <- read.csv("../data/raw/conflict.csv", header = TRUE)

```

2- Creating the binary armed conflict variable.

```

conflict_clean <- conflict |>
  mutate(
    armed_conflict = if_else(best >= 25, 1, 0)
    # binary variable indicating the presence of conflict for each country-year observation (0 =
  ) |>
  group_by(iso, year) |>
  summarise(
    armed_conflict = as.integer(max(armed_conflict)),
    .groups = "drop"
  )

conflict_clean <- conflict_clean |>
  mutate(
    year = as.integer(year+1)
  ) |>
  ungroup()

```

Merging all data sets

```

final_data <- list(
  matmor_long,
  infmor_long,
  neomor_long,
  un5mor_long,
  disaster_clean,
  conflict_clean
)

```

```
) |>
  reduce(full_join, by = c("iso", "year"))

write.csv(final_data, "../data/processed/final_data.csv", row.names=FALSE)
```